

Automatic Car Parking Space Management System



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Car parking lots in buildings are usually managed manually with manned kiosks that allow the cars into the parking lot based on the space available. These manual control systems are now giving way to automated car parking systems, which are cost-effective and also reduce waiting time for cars. Automated car parking systems are mostly controlled by a microcontroller.

Presented here is a microcontroller-based car parking space management system designed for a parking lot with parking capacity of a hundred cars. It does all the required work like automatic counting, and opening and closing of the gates without human intervention.

The boom barriers at the gates

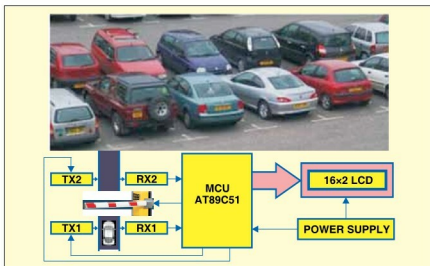


Fig. 1: Block diagram of automatic car parking space management system

can be powered by electric motors or hydraulic system controlled through a microcontroller. When the parking

space is full, the gate barrier does not open and cars cannot enter the parking lot.

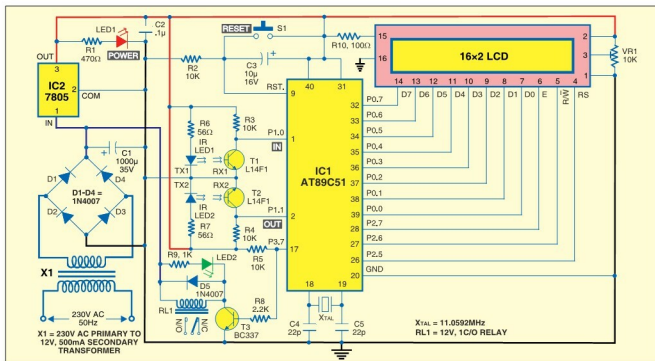


Fig. 2: Circuit for automatic car parking space management system